02-ROADMAP

LOT Pager — Roadmap & Risk Analysis

Phases (each phase ends with a pushed `.md` session report)

Phase	Weeks	Deliverable	Exit gate
P0 Plan (this session)	0	Doc set, BOM, decisions list	Inventor answers 5 blocking Qs
P1 Breadboard	1–4	ESP32-S3 dev kit + OLED/IPS + BME688 + camera + button, talking to staging `/api/device/*`	Notification shows; COPY appears in Log tab
P2 Server connector	2–5	`/api/device/pair`, `/poll`, `/event`, `device_*` log events, Profile → "Pair device"	Contract tests green; token revocation works
P3 EVT board	5–10	Custom 4-layer PCB (PCBWay, 5 pcs), firmware v0.1, 3D-printed shell	Boots, charges via Qi, 24 h battery
P4 Mechanical proto	8–14	5 CNC 316L sets, mirror polish, glass, gasket; RF + Qi through seam test	Wi-Fi RSSI > −70 dBm, Qi ≥ 70 % eff.
P5 DVT	14–20	20 units, `lot-link` flasher, test jig, PDF manuals v1	Drop/IP54 pass, OTA works
P6 PVT / 100-run	20–30	100 units via PCBWay	≥ 92 % first-pass yield
P7 Ship & learn	30+	Field telemetry, v1 shrink toward 5 mm	—

Critical path

Decisions → mechanical RF/Qi window design (P4 risk) → CNC lead time (stainless mirror polish 3–5 wks) → PVT. Start CNC quotes during P3.

Risk register

Risk	Likelihood	Impact	Mitigation
Steel kills Wi-Fi/Qi	High	Blocking	PEEK seam antenna slot; glass-side coil; test with coupons in P1
5 mm height impossible	Certain (full spec)	Scope	Stage 9 → 7 → 5 mm; accept thinner battery
Camera privacy/trust	Med	Brand	LED, stills-only, on-button-only, no background capture
Battery life (polling + screen)	High	UX	Light sleep, 60 s poll, screen off, wake on IMU tap/button; target 3 days @ 70 mAh
Mirror polish scratches/fingerprints	High	Quality	Optional PVD coating; ship cloth
Lithium cell shipping/certs	Med	Delay	UN38.3 cell, FCC/CE via module pre-cert (ESP32-S3-MINI is pre-certified)
Server cost of AI notifications	Low	$	Generate per user per day, cache, rate-limit device endpoints
Vendor lead times	Med	Delay	Order long-lead (CNC, glass) first

Compliance checklist (US-made brand)

FCC Part 15 (modular approval inherited if antenna untouched — a metal case near the antenna may void it → budget ~$5–8k for intentional-radiator test), UN38.3, RoHS. Estimates; verify with a lab.